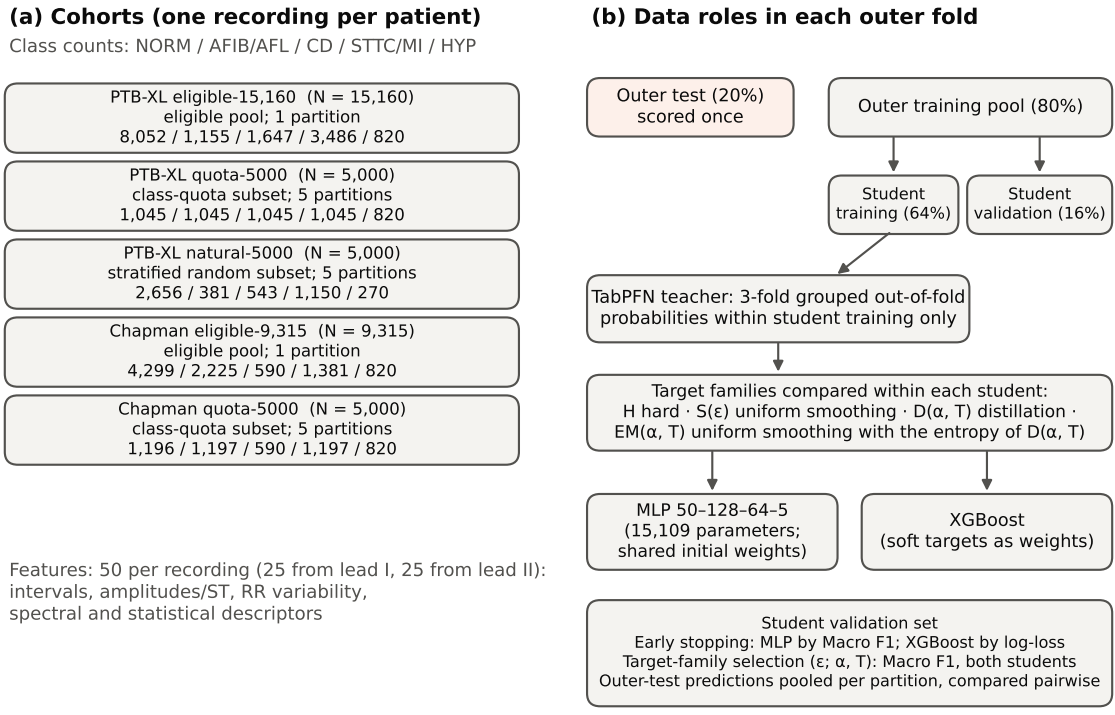
(a) Cohorts (one recording per patient)
Class counts: NORM / AFIB/AFL / CD / STTC/MI / HYP
PTB-XL eligible-15,160 (N = 15,160)
eligible pool; 1 partition
8,052 / 1,155 / 1,647 / 3,486 / 820
PTB-XL quota-5000 (N = 5,000)
class-quota subset; 5 partitions
1,045 / 1,045 / 1,045 / 1,045 / 820
PTB-XL natural-5000 (N = 5,000)
stratified random subset; 5 partitions
2,656 / 381 / 543 / 1,150 / 270
Chapman eligible-9,315 (N = 9,315)
eligible pool; 1 partition
4,299 / 2,225 / 590 / 1,381 / 820
Chapman quota-5000 (N = 5,000)
class-quota subset; 5 partitions
1,196 / 1,197 / 590 / 1,197 / 820
Features: 50 per recording (25 from lead I, 25 from lead II):
intervals, amplitudes/ST, RR variability,
spectral and statistical descriptors
(b) Data roles in each outer fold
Outer test (20%)
scored once
Outer training pool (80%)
Student
training (64%)
Student
validation (16%)
TabPFN teacher: 3-fold grouped out-of-fold
probabilities within student training only
Target families compared within each student:
H hard · S(ε) uniform smoothing · D(α, T) distillation ·
EM(α, T) uniform smoothing with the entropy of D(α, T)
MLP 50–128–64–5
(15,109 parameters;
shared initial weights)
XGBoost
(soft targets as weights)
Student validation set
Early stopping: MLP by Macro F1; XGBoost by log-loss
Target-family selection (ε; α, T): Macro F1, both students
Outer-test predictions pooled per partition, compared pairwise